# EXPERIMENT – 6

## Snort IDS/IPS to Detect and Prevent Real-Time Threats Using TryHackMe Platform

### Aim

To understand and demonstrate the working of Snort as an Intrusion Detection System (IDS) and Intrusion Prevention System (IPS) by detecting real-time network threats using the TryHackMe platform.

### Tools / Platform Required

- Web Browser (Chrome / Firefox)
- Internet Connection
- TryHackMe Account (Already Created)
- TryHackMe AttackBox (Built-in Linux Environment)
- Snort IDS/IPS Tool (Pre-installed in AttackBox)

### Theory (Brief)

Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS) are essential components of network security used to monitor, detect, and prevent malicious activities.
Snort is an open-source, signature-based IDS/IPS capable of real-time traffic analysis and packet inspection.
TryHackMe provides a safe and legal environment to practice IDS/IPS concepts by simulating real-world attacks and defenses without impacting live systems.

### Steps to Perform the Experiment

### Step 1: Login to TryHackMe

1. Open browser and go to: https://tryhackme.com
2. Login using registered credentials.

### Step 2: Open Snort Room

1. In the search bar, type **Snort**.
2. Select a room related to **Snort IDS/IPS**.
3. Click **Join Room**.

### Step 3: Start the AttackBox

1. Click **Start AttackBox**.
2. Wait for the Linux environment to load.
3. Open the terminal inside AttackBox.

### Step 4: Verify Snort Installation

1. In the terminal, check Snort version:
2. snort --version
3. Ensure Snort is successfully installed.

**Step 5: Run Snort in IDS Mode**

1. Start Snort in console mode:
2. sudo snort -A console -q -c /etc/snort/snort.conf -i eth0
3. Snort begins monitoring network traffic in real time.

**Step 6: Detect Network Threats**

1. Perform network scanning or simulated attacks as instructed in the TryHackMe room.
2. Observe alerts generated by Snort.
3. Analyze alerts indicating suspicious or malicious activity.

**Step 7: Configure Snort as IPS (Optional)**

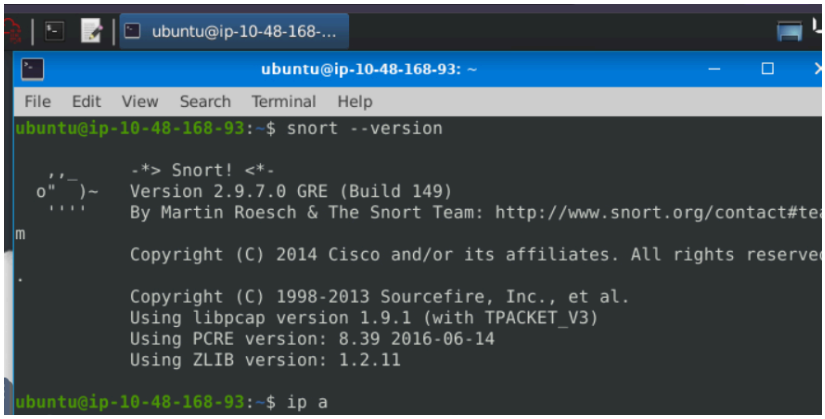
1. Run Snort in inline (IPS) mode:
2. sudo snort -Q --daq afpacket -c /etc/snort/snort.conf -i eth0
3. Configure rules to drop malicious packets.
4. Observe blocked traffic in real time.

**Step 8: Analyze Alerts**

1. Review generated alerts on the terminal.
2. Identify attack types such as port scanning, brute force, or suspicious packets.

**Output**

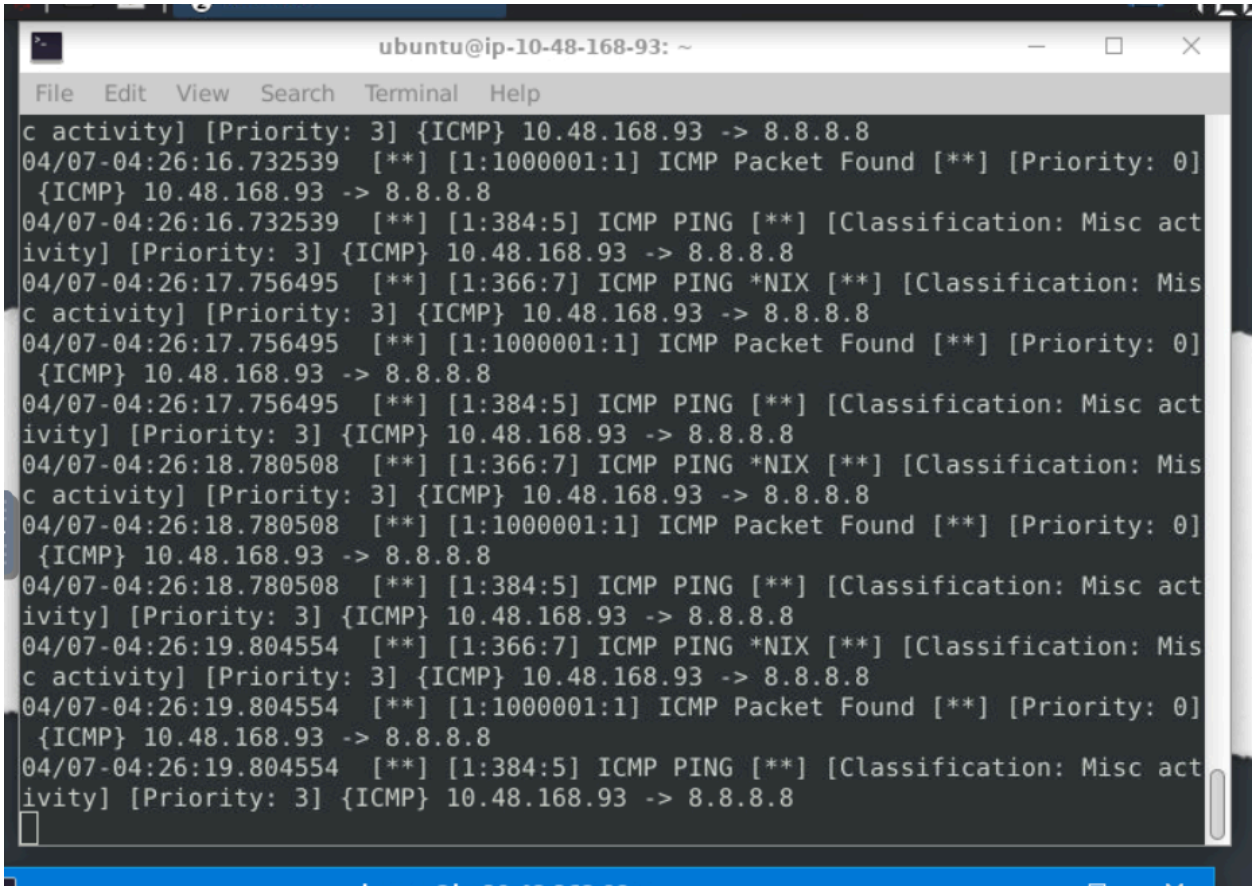
- Snort successfully detects malicious traffic.
- Real-time alerts are displayed in the terminal.
- Screenshots of alerts and completed tasks are recorded.



```
ubuntu@ip-10-48-168-93: ~
File  Edit  View  Search  Terminal  Help
           Copyright (C) 2014 Cisco and/or its affiliates. All rights reserved
.
           Copyright (C) 1998-2013 Sourcefire, Inc., et al.
           Using libpcap version 1.9.1 (with TPACKET_V3)
           Using PCRE version: 8.39 2016-06-14
           Using ZLIB version: 1.2.11

           Rules Engine: SF_SNORT_DETECTION_ENGINE  Version 2.4  <Build 1>
           Preprocessor Object: SF_POP  Version 1.0  <Build 1>
           Preprocessor Object: SF_SSH  Version 1.1  <Build 3>
           Preprocessor Object: SF_DNP3  Version 1.1  <Build 1>
           Preprocessor Object: SF_DNS  Version 1.1  <Build 4>
           Preprocessor Object: SF_GTP  Version 1.1  <Build 1>
           Preprocessor Object: SF_MODBUS  Version 1.1  <Build 1>
           Preprocessor Object: SF_SDF  Version 1.1  <Build 1>
           Preprocessor Object: SF_DCERPC2  Version 1.0  <Build 3>
           Preprocessor Object: SF_FTPTELNET  Version 1.2  <Build 13>
           Preprocessor Object: SF_REPUTATION  Version 1.1  <Build 1>
           Preprocessor Object: SF_SMTP  Version 1.1  <Build 9>
           Preprocessor Object: SF_IMAP  Version 1.0  <Build 1>
           Preprocessor Object: SF_SIP  Version 1.1  <Build 1>
           Preprocessor Object: SF_SSLPP  Version 1.1  <Build 4>
Commencing packet processing (pid=1692)
```

```
ubuntu@ip-10-48-168-93: ~
File  Edit  View  Search  Terminal  Help
buntu@ip-10-48-168-93:~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
```




**Result**

Thus, Snort IDS/IPS was successfully used to detect and prevent real-time network threats using the TryHackMe platform.

The experiment enhanced understanding of intrusion detection, prevention mechanisms, and real-time network security monitoring.